

Problem Set 3: Spontaneous Symmetry Breaking and other topics

1. *Derrick's theorem.* Consider a theory of N real scalar fields in $d + 1$ dimensional spacetime with standard kinetic terms and general potential V (assumed to be non-negative) described by

$$\mathcal{L} = \frac{1}{2} \sum_{j=1}^N \partial_\mu \phi_j \partial^\mu \phi_j - V(\phi_j).$$

Prove that for $d > 1$, there are no static, finite energy solutions to the field equations.

Hints:

- (a) Show that the equations satisfied by such a static field configuration, $\phi_j(\vec{x})$, can be derived by minimizing the energy.
 - (b) Write the total energy as a sum of gradient and potential energy, and analyze how it behaves under rescalings $\phi_j(\vec{x}) \rightarrow \phi'_j(\vec{x}) = \phi_j(\lambda \vec{x})$ with arbitrary λ .
 - (c) Show that for $d > 1$, the total energy has no static, finite energy, minimum.
2. *Linear sigma model.* Consider the following model in $d + 1$ spacetime dimensions

$$\mathcal{L} = \mathcal{L} = \frac{1}{2} \partial_\mu \vec{\phi}^T \partial^\mu \vec{\phi} + \frac{1}{2} m^2 \vec{\phi}^T \vec{\phi} - \frac{\lambda}{4} \left(\vec{\phi}^T \vec{\phi} \right)^2,$$

where $\vec{\phi} = (\phi_1, \dots, \phi_N)^T$.

- (a) Does this theory have a continuous symmetry? What is the vacuum manifold of the theory?

- (b) How many Goldstone bosons π_j can we expect? Show it explicitly by expanding the field around a particular choice of “true” vacuum $\vec{\phi}_0 = (0, \dots, 0, v)^T$ where v is a constant to be determined. How is this related to the Higgs mechanism?
- (c) For what choices of N and d does the model exhibit so-called topological defect solutions such as domain walls (sometimes called kinks) and strings (sometimes called vortices)? What kind of topological defects can we expect for $N = d$?
3. *A non-linear sigma model I.* Define the matrix field U as

$$U(x) = \exp \left(i \vec{\sigma} \cdot \vec{\phi}(x) \right),$$

where the σ^i are the Pauli matrices and let ϕ^i are three real scalar fields. What set does the matrix U belong to? Show that the theory with Lagrangian

$$\mathcal{L} = \frac{1}{2} \text{Tr} (\partial_\mu U^\dagger \partial^\mu U),$$

is invariant under the transformation $U(x) \rightarrow U'(x) = A U(x) B^\dagger$, where A and B are constant unitary matrices. Determine the form of the conserved currents that result from this symmetry.

4. *A non-linear sigma model II.* Denote the currents obtained in the previous problem by $j_k^\mu(x)$. Modify $\mathcal{L}$ by adding a “mass” term:

$$\mathcal{L} \rightarrow \mathcal{L} - \frac{\mu}{4} \text{Tr} (U + U^\dagger) + \mu.$$

Compute $\partial_\mu j_k^\mu(x)$ for fields satisfying the equations of motion derived from this modified Lagrangian, and determine which of the currents remain conserved. What is the symmetry they correspond to?

5. *An example of topological charge.* Let U be an $SU(N)$ -valued field.

- (a) Show that the continuity equation $\partial_\mu j^\mu$ holds for the current,

$$j^\mu = \epsilon^{\mu\nu\rho\sigma} \text{Tr} (U^\dagger \partial_\nu U U^\dagger \partial_\rho U U^\dagger \partial_\sigma U). \quad (1)$$

- (b) Show that the corresponding charge

$$Q = \int_{R^3} d^3x \epsilon^{ijk} \text{Tr} (U^\dagger \partial_i U U^\dagger \partial_j U U^\dagger \partial_k U),$$

is conserved.

Note that there is no need for the Euler–Lagrange equation and that we do not use Noether’s theorem.